# HackMISSO 2026 Technical Prompt

## Cyber Risk Assessment Platform

**Prompt Summary**

**Problem**:

Many small businesses, nonprofits, and student organizations rely on digital systems but **lack the resources to hire cybersecurity experts**. As a result, they often don't know:

- What their biggest cybersecurity risks are
- Which security improvements matter most
- How their practices compare to frameworks like **NIST, CIS Controls, or ISO 27001**

These frameworks are powerful but are **often complex for people who are non-technical**.

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**Your Challenge**

Build an **AI-powered cyber risk assessment platform** that helps organizations quickly understand their cybersecurity posture.

Your solution should allow a user to:

1. **Answer simple cybersecurity questions**
2. **Automatically generate a cyber risk score**
3. **Receive clear, prioritized security recommendations**

Your platform should use **AI and automation** to turn complex cybersecurity frameworks into **simple, actionable insights**.

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**Final Deliverables (these are required for your presentation):**

- **System Architecture Diagram:** shows components of system (frontend, backend, database, API, automation workflow)
- **Questionnaire and Scoring Matrix:** List of questions and a scoring strategy to show your understanding of an organization's security needs and what potential issues they may have

- **Pitch Presentation Slide Deck:** A short slide deck explaining your solution's vision, design, and potential impact. 6 slides max
- **UI/UX Mockup:** High-fidelity designs or a clickable prototype (e.g., using Figma) that shows what your application looks like and how a user would navigate it. Focus on the main portions of the product
- **AI Recommendation Agent:** Create a workable agent that extracts information from your questionnaire and gives recommendations

Important Notes:

- A production-ready system is **NOT** required. The main focus should be on the idea, logic, and prototyping. However, creating a **fully functional product** has the potential to benefit your final score
- Tools and technologies are largely up to the team's discretion unless specified

## ADDITIONAL INFORMATION

**Problems to Solve**

Your team must design a solution that addresses **at least two** of the following problems.

**Pain Point 1: Lack of Visibility Into Cyber Risk**

Organizations often don't know:

- What their biggest security weaknesses are
- Which risks are most urgent

Your solution could help identify and visualize **high-risk areas**.

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**Pain Point 2: Security Frameworks Are Too Complex**

Frameworks like:

- NIST Cybersecurity Framework
- CIS Critical Security Controls
- ISO 27001

are valuable but **overwhelming for non-experts**.

Your solution could translate these frameworks into **simple questions or checklists**.

Example:

Instead of:

"Implement identity access governance controls"

Ask:

"Do employees use multi-factor authentication when logging in?"

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**Pain Point 3: Organizations Struggle to Prioritize Security Improvements**

Most organizations don't know **what to fix first**.

Your platform should help answer questions like:

- “What is the most important thing we should fix this month?”
- “Which improvements reduce the most risk?”

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**Pain Point 4: Security Reports Are Hard to Understand**

Security reports are often filled with **technical jargon**.

Your solution should translate findings into **clear, actionable language**.

Example:

Instead of:

“Critical vulnerability in authentication mechanisms.”

Say:

“Your accounts do not use multi-factor authentication. If a password is stolen, attackers could access sensitive systems.”

## Deliverables (Detailed Requirements)

Each team must submit:

### 1️⃣ System Architecture Diagram

Create a visual diagram showing the major components of your system and how they interact. This diagram should illustrate the overall structure of your solution and how information flows through it.
Examples of components to include:

- Frontend interface (web app, form, chatbot, etc.)
- Backend logic or application layer
- Database or data storage
- External APIs or AI services
- Automation workflows (such as n8n)

The goal is to demonstrate **how data moves from user input to risk analysis and recommendations**.

### 2️⃣ Questionnaire and Scoring Matrix

Design the cybersecurity assessment used by your platform. This should include:

- The **questions** your platform asks users
- The **security category** each question belongs to (e.g., Access Control, Data Protection, Incident Response, etc.)
- A **scoring model** explaining how responses affect risk levels
- A brief explanation of how the **overall cybersecurity risk score is calculated**

This deliverable demonstrates your understanding of **how organizations evaluate cybersecurity posture and risk**.

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### 3️⃣ Pitch Presentation Slide Deck (Maximum 6 Slides)

Prepare a short presentation explaining your solution.

Your slides should clearly communicate:

1. **Problem** – What cybersecurity challenge are you addressing?
2. **Solution** – What platform did you design, and how does it help organizations?
3. **Technical Architecture** – How does your system work?
4. **Prototype / Demo** – Show the key features of your solution
5. **AI Agent** – Explain how your AI recommendation system works
6. **Impact** – Why does your solution matter, and who could benefit from it?

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### 4️⃣ UI/UX Mockup

Design a visual representation of your platform's interface.

This can be a **high-fidelity design or clickable prototype** created using tools such as **Figma**.

Your mockups should show the primary parts of your system, such as:

- The cybersecurity **assessment questionnaire**
- The **risk score dashboard or results page**
- The **AI-generated recommendations or report**

The focus should be on **clear user experience and usability**, not perfect visual design.

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### 5️⃣ AI Recommendation Agent

Create an AI-powered component that analyzes the results of your cybersecurity assessment and generates recommendations.

Your AI agent should be able to:

- Interpret responses from the questionnaire
- Identify **areas of high cybersecurity risk**
- Explain risks in **clear, easy-to-understand language**

- Provide **prioritized security recommendations** for improvement

This could be implemented using tools such as AI APIs, automation workflows, or chatbot interfaces.